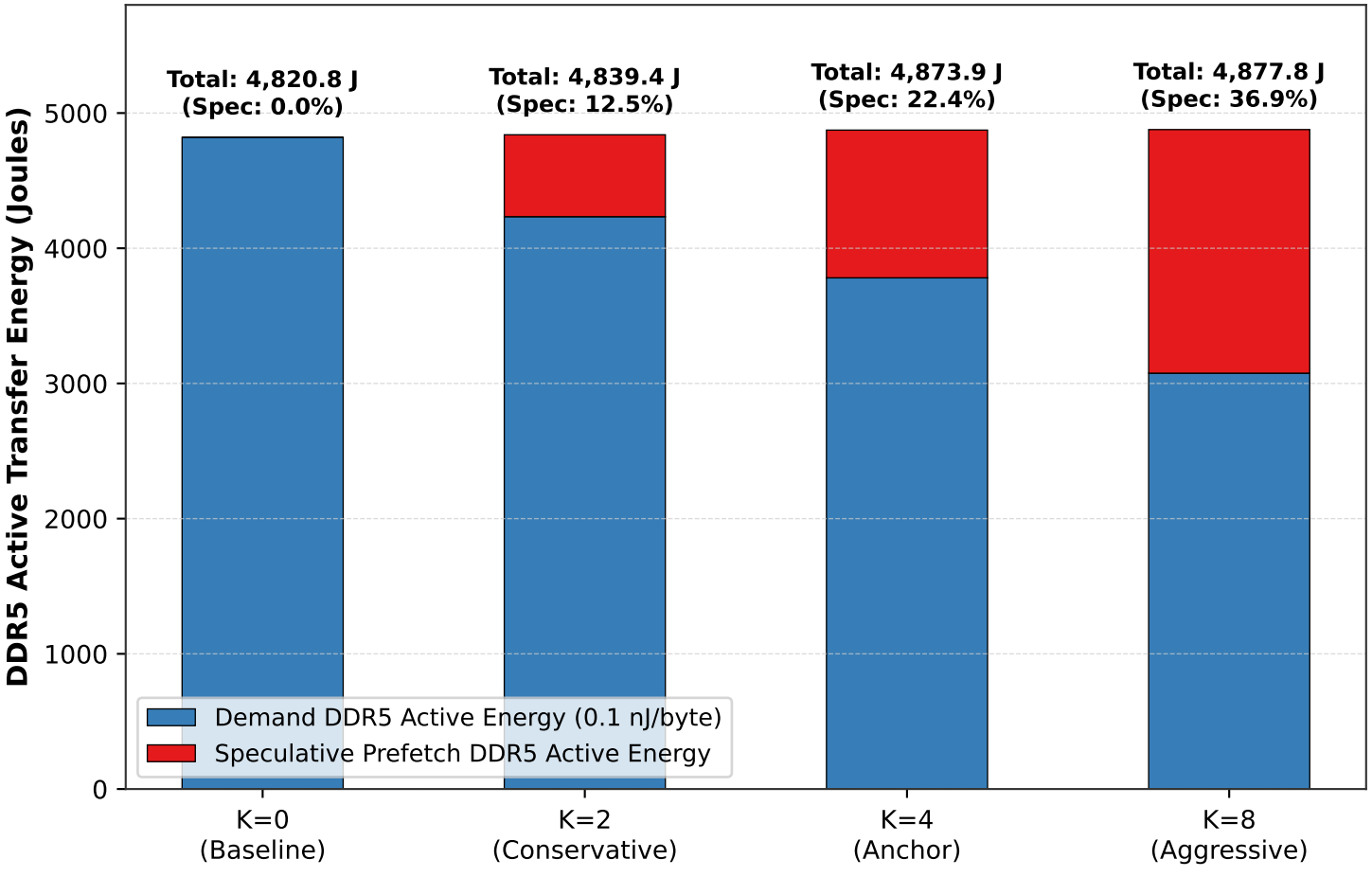
Figure 3: Demand vs. Speculative DDR5 Active Transfer Energy across Prefetch Budgets (B=32, P=2048, T=128, H=1)
DDR5 Active Transfer Energy (Joules)
0
1000
2000
3000
4000
5000
Total: 4,820.8 J
(Spec: 0.0%)
Total: 4,839.4 J
(Spec: 12.5%)
Total: 4,873.9 J
(Spec: 22.4%)
Total: 4,877.8 J
(Spec: 36.9%)
Demand DDR5 Active Energy (0.1 nJ/byte)
Speculative Prefetch DDR5 Active Energy
K=0
(Baseline)
K=2
(Conservative)
K=4
(Anchor)
K=8
(Aggressive)